

Table of Contents

Unit 1: Introduction to Computational Systems	2
Question Bank	2
Section A: Multiple Choice Questions	2
Topic: Theory of Systems	2
Topic: Software	4
Topic: Von Neumann Architecture	5
Topic: Number Systems	8
Topic: Data Representation	12
Topic: Binary Arithmetic	14
Topic: Text Encoding	16
Topic: Storage and Data Sizes	19
Section B: Short Questions	20
Section C: Long Questions	21

Unit 1: Introduction to Computational Systems

Question Bank

Section A: Multiple Choice Questions

Instructions: Choose the correct answer from the options given. The correct answer is marked with .

Topic: Theory of Systems

1. A system is best described as:

- a) A single component working alone
 - b) An organized collection of interdependent parts working together to achieve a common goal
 - c) A machine that processes only numbers
 - d) A program installed on a computer
-

2. The primary function of a system is:

- a) To work independently
 - b) To achieve a common goal
 - c) To create new systems
 - d) To provide entertainment
-

3. One of the fundamental concepts of any system is:

- a) Its size
 - b) Its objective
 - c) Its age
 - d) Its price
-

4. An example of a simple system is:

- a) A human body
 - b) A computer network
 - c) A thermostat regulating temperature
 - d) The Internet
-

5. The basic components of a system are:

- a) Users, hardware, software
 - b) Objectives, components, environment, communication
 - c) Inputs, outputs, processes
 - d) Sensors, actuators, controllers
-

6. The environment of a system refers to:

- a) The internal memory of a computer
 - b) The operating system installed on a device
 - c) Everything external to the system that interacts with it
 - d) The programming language used to build the system
-

7. Communication among system components is important because:

- a) It increases the cost of the system
 - b) It replaces the need for hardware
 - c) It ensures components work together to achieve the system's objectives
 - d) It reduces the number of components needed
-

8. An Information System is best described as:

- a) A single computer connected to the internet
 - b) A database of employee records
 - c) A combination of technology and people that collects, processes, stores, and shares data
 - d) A software program for managing files
-

Topic: Software

9. Software is defined as:

- a) Physical components of a computer
 - b) A collection of programs and instructions that tell a computer what to do
 - c) The screen and keyboard of a computer
 - d) The memory chips inside a computer
-

10. Which of the following is an example of system software?

- a) Microsoft Word
 - b) Google Chrome
 - c) Minecraft
 - d) Microsoft Windows
-

11. Which of the following is an example of application software?

- a) Microsoft Word
 - b) BIOS
 - c) Disk Cleanup
 - d) Device Manager
-

12. Which software is used to enhance system performance and security?

- a) Operating system
 - b) Utility software
 - c) Application software
 - d) Device drivers
-

13. Device drivers are a type of:

- a) Application software
 - b) System software
 - c) Programming language
 - d) Web browser
-

14. The main purpose of system software is to:

- a) Help users write documents
 - b) Play multimedia files
 - c) Manage hardware resources and provide a platform for application software
 - d) Browse the internet
-

15. Application software is typically:

- a) Pre-installed and cannot be removed
 - b) Installed by the user to perform specific tasks
 - c) Part of the CPU
 - d) Only used by programmers
-

16. Which of the following is NOT an example of system software?

- a) Operating system
 - b) Device driver
 - c) Web browser
 - d) Utility program
-

17. Which of the following is NOT an example of application software?

- a) Google Chrome
 - b) VLC Media Player
 - c) Fortnite
 - d) Linux
-

Topic: Von Neumann Architecture

18. Which of the following describes the Von Neumann architecture's main characteristic?

- a) Separate memory for data and instructions
 - b) Parallel execution of instructions
 - c) Single memory store for both program instructions and data
 - d) Multiple CPUs for different tasks
-

19. The Von Neumann architecture was developed during the:

- a) 1920s
 - b) 1930s
 - c) 1940s
 - d) 1960s
-

20. Which component of the CPU performs mathematical and logical calculations?

- a) Control Unit (CU)
 - b) Program Counter (PC)
 - c) Arithmetic Logic Unit (ALU)
 - d) Instruction Register (IR)
-

21. The Control Unit (CU) is responsible for:

- a) Storing all data permanently
 - b) Governing the activities of the CPU and coordinating its components
 - c) Displaying output on the monitor
 - d) Connecting the computer to the internet
-

22. Which component stores the memory address of the next instruction to be executed?

- a) Instruction Register (IR)
 - b) Arithmetic Logic Unit (ALU)
 - c) Program Counter (PC)
 - d) Control Bus
-

23. The correct order of the Von Neumann instruction cycle is:

- a) Execute → Fetch → Decode → Store
 - b) Decode → Fetch → Execute → Store
 - c) Fetch → Decode → Execute → Store
 - d) Store → Fetch → Decode → Execute
-

■

24. During the Fetch stage of the Von Neumann cycle, the CPU:

- a) Performs the calculation
 - b) Retrieves an instruction from memory
 - c) Sends output to the monitor
 - d) Decodes the operation code
-

■

25. The System Bus is used to:

- a) Power the CPU
 - b) Store data permanently
 - c) Facilitate the movement of data between components inside a computer
 - d) Connect the computer to the internet
-

■

26. Which bus carries the actual data being transferred between components?

- a) Address Bus
 - b) Control Bus
 - c) Data Bus
 - d) System Bus
-

■

27. Which bus carries information about the destination of data?

- a) Data Bus
 - b) Address Bus
 - c) Control Bus
 - d) Memory Bus
-

■

28. RAM is an example of which Von Neumann component?

- a) Input device
 - b) Output device
 - c) CPU
 - d) Memory
-

29. Which of the following is an input device?

- a) Monitor
 - b) Printer
 - c) Speaker
 - d) Microphone
-

30. Which of the following is an output device?

- a) Keyboard
 - b) Mouse
 - c) Monitor
 - d) Scanner
-

Topic: Number Systems

31. The decimal number system has a base of:

- a) 2
 - b) 8
 - c) 10
 - d) 16
-

32. The binary number system has a base of:

- a) 2
 - b) 8
 - c) 10
 - d) 16
-

33. The octal number system uses digits from:

- a) 0 to 6
 - b) 0 to 7
 - c) 0 to 8
 - d) 0 to 9
-

34. The hexadecimal number system has a base of:

- a) 2
 - b) 8
 - c) 10
 - d) 16
-

35. In the hexadecimal system, the letter 'A' represents the decimal value:

- a) 9
 - b) 10
 - c) 11
 - d) 12
-

36. In the hexadecimal system, the letter 'F' represents the decimal value:

- a) 14
 - b) 15
 - c) 16
 - d) 17
-

37. The binary number 1011 in decimal is:

- a) 9
 - b) 10
 - c) 11
 - d) 12
-

38. The decimal number 83 in binary is:

- a) 1001011
 - b) 1010011
 - c) 1100011
 - d) 1011001
-

■

39. The decimal number 83 in octal is:

- a) 112
- b) 123
- c) 132
- d) 143

■

40. Each octal digit is represented by how many binary bits?

- a) 2 bits
- b) 3 bits
- c) 4 bits
- d) 8 bits

■

41. Each hexadecimal digit is represented by how many binary bits?

- a) 2 bits
- b) 3 bits
- c) 4 bits
- d) 8 bits

■

42. The binary number 110101011 in octal is:

- a) 523
- b) 635
- c) 653
- d) 563

■

43. The binary number 1101011010110010 in hexadecimal is:

- a) D2B6
- b) 6BD2
- c) B6D2
- d) D6B2

■

44. The hexadecimal number 1A3 in decimal is:

- a) 413
 - b) 429
 - c) 419
 - d) 439
-

45. When converting decimal to binary, you divide by:

- a) 8
 - b) 10
 - c) 16
 - d) 2
-

46. When converting decimal to octal, you divide by:

- a) 2
 - b) 8
 - c) 10
 - d) 16
-

47. When converting decimal to hexadecimal, you divide by:

- a) 2
 - b) 8
 - c) 10
 - d) 16
-

48. In decimal-to-binary conversion, the binary result is obtained by reading the remainders:

- a) From top to bottom
 - b) Left to right
 - c) From bottom to top
 - d) In any order
-

49. Why do computers use the binary system?

- a) Binary is easier for humans to read
 - b) Binary has more digits than decimal
 - c) Binary maps directly to the two states of electronic circuits (ON and OFF)
 - d) Binary was invented before decimal
-

50. The octal number system was commonly used in:

- a) Modern smartphones
 - b) Early computing systems like the PDP-8
 - c) Modern supercomputers
 - d) Cloud storage systems
-

Topic: Data Representation

51. A 1-byte unsigned integer can store a maximum value of:

- a) 127
 - b) 255
 - c) 512
 - d) 1024
-

52. The maximum value that can be stored in n bits (unsigned) is:

- a) $2n$
 - b) $2n + 1$
 - c) $2^n - 1$
 - d) $2n - 2$
-

53. In a signed integer, which bit is used to indicate a negative number?

- a) The least significant bit (rightmost bit)
 - b) The most significant bit (leftmost bit)
 - c) The middle bit
 - d) All bits equally
-

■

54. A 1-byte signed integer can store a maximum positive value of:

- a) 255
- b) 127
- c) 128
- d) 256

■

55. The minimum value of a 1-byte signed integer is:

- a) -127
- b) -128
- c) -255
- d) -256

■

56. 1's Complement is obtained by:

- a) Adding 1 to the binary number
- b) Dividing by 2
- c) Inverting all bits (changing 0s to 1s and 1s to 0s)
- d) Multiplying by -1

■

57. 2's Complement is obtained by:

- a) Inverting all bits only
- b) Adding 2 to the binary number
- c) Inverting all bits and then adding 1 to the LSB
- d) Subtracting 1 from the binary number

■

58. The 8-bit 2's complement representation of -5 is:

- a) 11111010
- b) 11111011
- c) 10000101
- d) 00000101

59. How many bytes does a typical 4-byte integer use?

- a) 1 byte
 - b) 2 bytes
 - c) 4 bytes
 - d) 8 bytes
-

60. The maximum value of a 2-byte unsigned integer is:

- a) 255
 - b) 32,767
 - c) 65,535
 - d) 4,294,967,295
-

Topic: Binary Arithmetic

61. In binary addition, $1 + 1$ equals:

- a) 2
 - b) 10 in decimal
 - c) 0 with a carry of 1
 - d) 11
-

62. In binary addition, $1 + 1 + 1$ (with carry) equals:

- a) 0 with carry 1
 - b) 1 with carry 1
 - c) 11
 - d) 10
-

63. The result of binary addition $1101 + 1011$ is:

- a) 10110
 - b) 11100
 - c) 11000
 - d) 10100
-

64. Binary subtraction is most commonly performed using:

- a) 1's complement only
 - b) Direct subtraction
 - c) 2's complement addition
 - d) Hexadecimal conversion
-

65. When subtracting using 2's complement, after adding the numbers, any carry beyond the word size is:

- a) Added back to the result
 - b) Discarded
 - c) Stored in the sign bit
 - d) Used as the final answer
-

66. The result of $9 - 6$ in binary using 2's complement is:

- a) 0010
 - b) 0011
 - c) 0100
 - d) 0101
-

67. In binary multiplication, 101×11 equals:

- a) 1100
 - b) 1110
 - c) 1111
 - d) 10001
-

68. In binary division, $1100 \div 10$ equals:

- a) 101
 - b) 110
 - c) 111
 - d) 100
-

Topic: Text Encoding

69. ASCII stands for:

- a) Advanced Standard Code for Information Interchange
 - b) American Standard Communication for Information Interchange
 - c) American Standard Code for Information Interchange
 - d) Advanced Standard Communication for Information Interchange
-

70. How many bits are used in the standard ASCII encoding?

- a) 7 bits
 - b) 8 bits
 - c) 16 bits
 - d) 32 bits
-

71. How many characters can standard ASCII represent?

- a) 64
 - b) 128
 - c) 256
 - d) 512
-

72. Extended ASCII uses how many bits?

- a) 7 bits
 - b) 8 bits
 - c) 16 bits
 - d) 32 bits
-

73. Extended ASCII can represent how many characters?

- a) 128
 - b) 256
 - c) 512
 - d) 1024
-

74. The ASCII code for the uppercase letter 'P' is:

- a) 70
 - b) 75
 - c) 80
 - d) 85
-

75. The ASCII code for the lowercase letter 'a' is:

- a) 65
 - b) 97
 - c) 107
 - d) 115
-

76. Unicode was developed to:

- a) Replace binary representation
 - b) Map all graphic characters from all the world's writing systems
 - c) Improve CPU speed
 - d) Reduce file sizes
-

77. UTF stands for:

- a) Universal Text Format
 - b) Unified Transfer Format
 - c) Unicode Transformation Format
 - d) Universal Transfer File
-

78. UTF-8 is:

- a) A fixed-length encoding using 8 bytes
 - b) A variable-length encoding using 1 to 4 bytes per character
 - c) Compatible only with Unicode
 - d) The same as ASCII
-

79. UTF-8 is backward compatible with:

- a) UTF-16
 - b) UTF-32
 - c) ASCII
 - d) Unicode only
-

80. UTF-16 uses how many bytes per character?

- a) 1 byte only
 - b) 4 bytes only
 - c) Either 2 bytes or 4 bytes
 - d) 1 to 4 bytes
-

81. UTF-32 uses a fixed size of:

- a) 1 byte per character
 - b) 2 bytes per character
 - c) 4 bytes per character
 - d) 8 bytes per character
-

82. The Urdu letter 'ب' in UTF-8 occupies:

- a) 1 byte
 - b) 2 bytes
 - c) 3 bytes
 - d) 4 bytes
-

83. The letter 'A' in Unicode is represented as:

- a) U+0041 in hex, 8 bits in binary
 - b) U+0041, which is 01000001 in binary (1 byte in UTF-8)
 - c) U+0065 in hex
 - d) U+0100, which is 2 bytes in UTF-8
-

Topic: Storage and Data Sizes

84. How many bits are in 1 byte?

- a) 4
 - b) 8
 - c) 16
 - d) 32
-

85. 1 Kilobyte (KB) is equal to:

- a) 1000 bytes
 - b) 1024 bytes
 - c) 512 bytes
 - d) 2048 bytes
-

86. 1 Gigabyte (GB) is equal to:

- a) 1024 Bytes
 - b) 1024 Kilobytes
 - c) 1024 Megabytes
 - d) 1024 Terabytes
-

87. Which storage device uses spinning disks to read and write data?

- a) SSD
 - b) USB Flash Drive
 - c) HDD (Hard Disk Drive)
 - d) Cloud Storage
-

88. Which storage device uses flash memory for faster access?

- a) HDD
 - b) SSD (Solid State Drive)
 - c) CD-ROM
 - d) Magnetic Tape
-

89. All data stored in a computer is ultimately represented as:

- a) Decimal numbers
 - b) Text characters
 - c) Hexadecimal codes
 - d) Binary digits (0s and 1s)
-

90. The unit larger than a Terabyte is:

- a) Gigabyte
 - b) Megabyte
 - c) Kilobyte
 - d) Petabyte
-
-

Section B: Short Questions

Instructions: Answer each question in 2–4 sentences.

1. Define a system.
2. What is an Information System?
3. What is the role of the environment in a system?
4. Why is communication among system components important?
5. What is software?
6. Define system software and provide two examples.
7. Define application software and provide two examples.
8. Differentiate between system software and application software.
9. Enlist the main components of the Von Neumann architecture.
10. Enlist the four main steps in the Von Neumann instruction cycle.
11. What is the role of the ALU in a computer system?
12. What is the role of the Control Unit (CU)?
13. What is a System Bus? Name its three types.
14. What is a key advantage of the Von Neumann architecture?
15. What is the decimal number system? Give an example of a positional value calculation.
16. Why do computers use the binary number system?

17. What is the octal number system? What is its base?
 18. What is the hexadecimal number system? What digits does it use?
 19. How many binary bits represent one octal digit? Why?
 20. How many binary bits represent one hexadecimal digit? Why?
 21. What are whole numbers? Give two examples of how computers use them.
 22. What are signed integers? How is the sign bit used?
 23. What is 1's Complement? How is it calculated?
 24. What is 2's Complement? Write the steps to find it.
 25. How does the number of bits affect the range of integer values?
 26. What is the primary purpose of the ASCII encoding scheme?
 27. How many characters does standard ASCII represent? How many bits does it use?
 28. What is Extended ASCII? How does it differ from standard ASCII?
 29. Define Unicode.
 30. What does UTF stand for? Name three UTF encoding types.
 31. What is UTF-8? What makes it useful?
 32. How is UTF-16 different from UTF-8?
 33. What is UTF-32? What is its main drawback?
 34. State the four rules of binary addition.
 35. Add 1100_2 and 1011_2 . Show your working.
 36. Subtract 0011_2 from 1010_2 using 2's complement. Show your working.
 37. List the steps of binary multiplication.
 38. List the steps of binary division.
 39. Name three types of storage devices used in computers.
 40. What is the difference between an HDD and an SSD?
 41. Write the data size hierarchy from Bit to Terabyte.
 42. What is the ASCII code for 'a', 'A', and space? Why are they different?
-

Section C: Long Questions

Instructions: Answer each question in detail with examples where applicable.

1. Describe the basic concepts of a system, including its objective, components, environment, and communication.

2. Design a simple real-world system of your choice. Identify and explain its objective, components, interactions between components, and its environment.
 3. Explain the difference between system software and application software. Include at least three examples of each and describe their roles within a computer system.
 4. Explain the Von Neumann Architecture of a computer. Describe all its major components and their roles.
 5. Explain the working of the Von Neumann Architecture. Describe all four stages of the instruction cycle (Fetch, Decode, Execute, Store) with a practical example.
 6. Explain the four number systems — Decimal, Binary, Octal, and Hexadecimal — including their base values, digits used, and an example of positional value calculation for each.
 7. Demonstrate the conversion of the decimal number 156 into:
 - (a) Binary
 - (b) Octal
 - (c) HexadecimalShow all working steps clearly.
 8. Explain how whole numbers and signed integers are stored in computer memory. Include the concept of the sign bit and the range of values for 1-byte and 2-byte representations.
 9. Explain the concept of 2's Complement. Why do computers use it to represent negative numbers? Demonstrate with the example of -13 in 8-bit binary.
 10. Explain how characters are encoded using ASCII and Unicode. Provide examples from both English and Urdu. Also explain the difference between UTF-8, UTF-16, and UTF-32.
 11. Perform the following binary arithmetic operations with full working steps:
 - (a) Addition: $11011_2 + 10101_2$
 - (b) Subtraction: $10110_2 - 01011_2$ (using 2's complement)
 - (c) Multiplication: $101_2 \times 11_2$
 - (d) Division: $1100_2 \div 10_2$
 12. Describe how different types of data — text, images, audio, and video — are ultimately stored as binary in a computer. Include the role of encoding schemes and storage devices.
-

End of Unit 1 Question Bank